

Chemistry marking guide and response

External assessment 2022

Combination response (100 marks)

Assessment objectives

This assessment instrument is used to determine student achievement in the following objectives:

1. describe and explain chemical equilibrium systems, oxidation and reduction, properties and structure of organic materials, and chemical synthesis and design
2. apply understanding of chemical equilibrium systems, oxidation and reduction, properties and structure of organic materials, and chemical synthesis and design
3. analyse evidence about chemical equilibrium systems, oxidation and reduction, properties and structure of organic materials, and chemical synthesis and design to identify trends, patterns, relationships, limitations or uncertainty
4. interpret evidence about chemical equilibrium systems, oxidation and reduction, properties and structure of organic materials, and chemical synthesis and design to draw conclusions based on analysis.

Note: Objectives 5, 6 and 7 are not assessed in this instrument.

Purpose

This document consists of a marking guide and a sample response.

The marking guide:

- provides a tool for calibrating external assessment markers to ensure reliability of results
- indicates the correlation, for each question, between mark allocation and qualities at each level of the mark range
- informs schools and students about how marks are matched to qualities in student responses.

The sample response:

- demonstrates the qualities of a high-level response
- has been annotated using the EAMG.

Mark allocation

Where a response does not meet any of the descriptors for a question or a criterion, a mark of '0' will be recorded.

Where no response to a question has been made, a mark of 'N' will be recorded.

Allowing for FT error — refers to 'follow through', where an error in the prior section of working is used later in the response, a mark (or marks) for the rest of the response can be awarded so long as it still demonstrates the correct conceptual understanding or skill in the rest of the response.

Marking guide

Multiple choice

Question	Response
1	A
2	D
3	B
4	B
5	D
6	A
7	D
8	D
9	A
10	B
11	D
12	D
13	C
14	C
15	A
16	C
17	C
18	C
19	A
20	A

Paper 1: Short response

Q	Sample response	The response:	Notes
21a)	2-bromopropane is saturated because it contains only single bonds.	- identifies 2-bromopropane is saturated [1 mark] - indicates 2-bromopropane contains only single bonds [1 mark]	
21b)	2-bromopropane is a secondary halogenoalkane, because the carbon that the bromine is attached to two carbon atoms.	- determines 2-bromopropane is a secondary halogenoalkane [1 mark] - explains that the bromine (halogen) is bonded to a carbon that is attached to two other carbon atoms [1 mark]	For the explanation, accept a structural formula of 2-bromopropane.

Q	Sample response	The response:	Notes
22	$[\text{H}_3\text{O}^+] = 10^{-4} \text{ and } [\text{F}^-] = 10^{-4}$ $[\text{HF}] = \frac{[\text{H}_3\text{O}^+][\text{F}^-]}{K_a}$ $[\text{HF}] = \frac{10^{-4} \times 10^{-4}}{7.2 \times 10^{-4}}$ $[\text{HF}] = 1.4 \times 10^{-5} \text{ mol L}^{-1}$	• uses correct substitution [1 mark] • calculates $[\text{HF}] = 1.4 \times 10^{-5}$ [1 mark]	Allow FT error for $[\text{HF}] = 2.2 \times 10^5 \text{ mol L}^{-1}$ if $[\text{H}_3\text{O}^+] = 4$ and $[\text{F}^-] = 4$ are used. Do not penalise incorrect significant figures.
23	Process 1: atom economy = $206.0/514.5 \times 100$ = 40.04% Process 2: atom economy = $206.0/266.0 \times 100$ = 77.44% Process 2 has 37.4% better atom economy than process 1 Economic impact: Process 2 has a better atom economy than process 1 (fewer reagents are required). Environmental impact: Process 2 is greener than process 1 because fewer waste products (atoms) are produced.	• calculates atom economy for – Process 1 as 40% [1 mark] – Process 2 as 77% [1 mark] • concludes process 2 is – cheaper as fewer reagent atoms are required [1 mark] – greener as fewer waste atoms are produced [1 mark]	For <i>cheaper</i>, accept: – fewer reagents are required – better atom economy. Allow FT error for conclusions correctly based on incorrect calculations.

Q	Sample response	The response:	Notes
24a)	Similarity: Both the Pt and Cu half-cells have a positive standard reduction potential compared to SHE. Difference: Pt half-cell is more positive than Cu half-cell. Significance: Cu electrode is oxidised (loses electrons) and the $\text{Fe}^{3+}(\text{aq})$ is reduced.	- Similarities: identifies that both half-cells have positive standard reduction potential [1 mark] - Differences: identifies that Pt half-cell is more positive than Cu half-cell [1 mark] - Significance: explains that Cu electrode is oxidised and Fe^{3+} is reduced [1 mark]	For <i>Cu electrode</i>, accept – Cu metal – Cu in Cu half-cell. Do not accept Fe (iron) is reduced.
24b)	Balance redox equation: $2\text{Fe}^{3+} + \text{Cu} \rightarrow \text{Cu}^{2+} + 2\text{Fe}^{2+}$	provides correct balanced redox equation [1 mark]	Allow FT error based on incorrect half-equations.
24c)	Cell potential = $0.77 + (-0.34) = +0.43 \text{ V}$	determines cell potential is +0.43 [1 mark]	Reduction half-equation: $\text{Fe}^{3+} + \text{e}^{-} \rightarrow \text{Fe}^{2+} = +0.77 \text{ V}$ Oxidation half-equation: $\text{Cu} \rightarrow \text{Cu}^{2+} + 2\text{e}^{-} = -0.34 \text{ V}$
24d)	Fe^{3+} is the oxidising agent, because ON decreases from +3 to +2.	- identifies Fe^{3+} as the oxidising agent [1 mark] - indicates oxidation number decreases [1 mark]	Do not accept Fe is reduced. For <i>oxidation number decreases</i>, accept <i>an electron is gained</i>.

Q	Sample response	The response:	Notes
25a)	The reaction occurred in a closed system, therefore matter is not exchanged with the surroundings. This means the system can reach a steady state, where the rate of the forward reaction (conversion of reactants to products) equals the rate of the reverse reaction (products are converted back into reactants) to establish a state of dynamic equilibrium.	- identifies gases reach dynamic equilibrium [1 mark] - explains matter is not exchanged with surrounds [1 mark] - explains rate of forward reaction is equal to rate of reverse reaction [1 mark]	For <i>matter is not exchanged</i>, accept <i>products are converted back into reactants</i>.
25b)	$K_c = \frac{[XA_3]^2}{[A_2]^3 [X_2]}$ $K_c = \frac{[4.2]^2}{[3.4]^3 [1.8]}$ $K_c = \frac{17.64}{70.7472} = 0.25$ $K_c < 1$ therefore equilibrium lies to the left	- determines $K_c = 0.25$ [1 mark] - indicates that since $K_c < 1$, equilibrium lies towards the reactants [1 mark]	For <i>reactants</i>, accept <i>left</i>. For $K_c < 1$, accept <i>[reactants] > [products]</i>. Allow FT error for equilibrium position if the K_c is correctly calculated from an incorrect expression.

Q	Sample response	The response:	Notes
26a)	pH = 4.0	determines pH for solution A is 4.0 [1 mark]	Do not penalise incorrect decimal places.
26b)	[H ⁺] in Solution B = 0.010 mol L ⁻¹	determines [H⁺] for solution B is 0.01 [1 mark]	Do not penalise incorrect significant figures. Also accept 1.0×10^{-2} and 10^{-2} .
26c)	pH = $-\log [0.063] = 1.2$ pOH = $14 - 1.2 = 12.8$	- provides relevant working [1 mark] - calculates pOH = 12.8 [1 mark]	Do not penalise incorrect decimal places. Relevant working may include: - pH = $-\log [0.063]$ $[\text{OH}^-] = \frac{1.00 \times 10^{-14}}{6.3 \times 10^{-2}}$ - pOH = $-\log [1.6 \times 10^{-13}]$ - or other suitable response consistent with a reasonable understanding.
27	Solutions A, C and D are acids and solutions B and E are bases, because phenol red is yellow when pH < 6.8 and red when pH > 8.4. Solution C is a weak electrolyte (from low conductivity), therefore C is propanoic acid. Solution D has higher conductivity than solution A, therefore D is sulfuric acid and solution A is HCl, because H₂SO₄ is diprotic acid and will give a higher concentration of ions in solution than monoprotic HCl. Solution E is KOH as it has a higher conductivity and therefore is a strong base. Solution B has a lower conductivity and therefore is ammonia, a weak base.	- identifies all five solutions [1 mark] - uses indicator data to identify acids and bases [1 mark] - uses conductivity data to identify relative strength of bases [1 mark] - uses conductivity data to identify relative strength of acids [1 mark] - identifies the diprotic acid is more conductive than monoprotic acid [1 mark]	A = HCl B = NH₃ C = CH₃CH₂COOH D = H₂SO₄ E = KOH

Paper 2: Short response

Q	Sample response	The response:	Notes
1a)	Compound A is an alcohol.	identifies the class as alcohol [1 mark]	
1b) i)	$ \begin{array}{ccccccc} & \text{H} & \text{H} & \text{H} & \text{H} & & \\ & & & & & & \\ \text{H} & - \text{C} & - \text{C} & - \text{C} & - \text{C} & - \text{O} & - \text{H} \\ & & & & & & \\ & \text{H} & \text{H} & \text{H} & \text{H} & & \end{array} $ Compound A is butanol.	- deduces structural formula for butanol [1 mark] - provides IUPAC name [1 mark]	For IUPAC name, accept: – butanol – 1-butanol – butan-1-ol Allow FT mark for correct IUPAC name based on incorrect structural formula for compound A.
1b) ii)	$ \begin{array}{ccccccc} \text{CH}_3 & - & \text{CH} & - & \text{CH}_2 & - & \text{CH}_3 \\ & & & & & & \\ & & \text{OH} & & & & \end{array} $ Compound C is 2-butanol.	- deduces structural formula for 2-butanol [1 mark] - provides IUPAC name [1 mark]	For IUPAC name, accept: – 2-butanol – butan-2-ol Allow FT mark for correct IUPAC name based on incorrect structural formula for compound C.
1c)	Butanol and 2-butanol are structural isomers.	identifies structural isomers [1 mark]	Allow FT error from 1b).
1d)	$ \begin{array}{ccccccc} & \text{H} & & \text{O} & & & \\ & & & & & & \\ \text{H} & - \text{C} & - & \text{C} & & & \\ & & & \diagdown & & & \\ & \text{H} & & \text{O} & - & \text{C} & - \text{C} & - \text{C} & - \text{C} & - \text{H} \\ & & & & & & & & & \\ & & & & & \text{H} & \text{H} & \text{H} & \text{H} & \end{array} $ Compound F is butyl ethanoate.	- provides structural formula for butyl ethanoate [1 mark] - provides IUPAC name as butyl ethanoate [1 mark]	Allow FT error for correct IUPAC name based on incorrect structural formula for compound F.

Q	Sample response	The response:	Notes
2a)	Exothermic Increasing temperature decreases K_c , indicating that the equilibrium shifts towards the reactants (endothermic) direction.	- determines forward reaction is exothermic [1 mark] - explain that the decrease in K_c as temperature increases indicates endothermic direction is towards the reactants [1 mark]	
2b)	$8.61 \times 10^{11} = \frac{[\text{SO}_3]^2}{(0.860)^2 (0.330)}$ $[\text{SO}_3]^2 = (0.7396)(0.330)(8.61 \times 10^{11}) = 2.10 \times 10^{11}$ Concentration = $4.58 \times 10^5 \text{ M}$	- provides appropriate working [1 mark] - calculates $[\text{SO}_3] = 4.58 \times 10^5$ [1 mark]	Do not penalise incorrect significant figures.
2c)	Halving the volume would double the pressure. To reduce the pressure, the equilibrium would shift toward the product to reduce the number of molecules present. However, due to the reactant decreasing the equilibrium constant remains unchanged.	- indicates that halving the volume doubles the pressure [1 mark] - explains that equilibrium will shift to reduce the number of molecules present to reduce pressure [1 mark] - explains that equilibrium will shift toward the product [1 mark] - explains why the equilibrium constant would not change [1 mark]	For <i>why the equilibrium constant would not change</i> , accept - constant temperature - corresponding decrease in reactants.

Q	Sample response	The response:	Notes
3a)	$\text{CH}_3\text{COOH}(\text{aq}) + \text{OH}^-(\text{aq}) \rightleftharpoons \text{CH}_3\text{COO}^-(\text{aq}) + \text{H}_2\text{O}(\text{l})$ The conjugate base formed is CH_3COO^- .	- provides correct balanced chemical equation [1 mark] - identifies CH_3COO^- as conjugate base [1 mark]	The balanced equation must include $\rightleftharpoons$ to be correct. For the balanced equation, accept: $\text{CH}_3\text{COOH} + \text{NaOH} \rightleftharpoons \text{CH}_3\text{COONa} + \text{H}_2\text{O}$
3b)	$K_b = \frac{1.00 \times 10^{-14}}{10^{-4.76}} = 10^{-9.24} = 5.75 \times 10^{-10}$	determines K_b is 5.75×10^{-10}	Do not penalise incorrect decimal places. Also accept $10^{-9.24}$
3c)	At equivalence point, moles OH^- = moles H^+ = moles CH_3COO^- $\text{moles CH}_3\text{COO}^- = n \times V = 0.10 \times 0.015 = 1.50 \times 10^{-3}$	- determines moles CH_3COO^- is 1.50×10^{-3} [1 mark] - calculates $[\text{CH}_3\text{COO}^-]$ is $2.31 \times 10^{-2} \text{ mol L}^{-1}$ [1 mark]	Do not penalise incorrect significant figures. Allow FT error from incorrect moles CH_3COO^- .
3d)	$[\text{CH}_3\text{COOH}] = [\text{OH}^-] = x$ $K_b = \frac{[\text{OH}^-][\text{CH}_3\text{COOH}]}{[\text{CH}_3\text{COO}^-]}$ $K_b = 5.75 \times 10^{-10} = \frac{x^2}{[2.31 \times 10^{-2}]}$ $x^2 = 1.33 \times 10^{-11}$ $x = 3.64 \times 10^{-6} = [\text{OH}^-]$ $\text{pOH} = -\log(3.64 \times 10^{-6}) = 5.44 \sim 5.4$ $\text{pH} = 14 - 5.4 = 8.6$	- provides correct substitution [1 mark] - calculates $[\text{OH}^-]$ is $3.64 \times 10^{-6} \text{ M}$ [1 mark] - determines pOH is 5.4 [1 mark] - calculates pH is 8.6 [1 mark]	Allow FT error from: – 3b), i.e. K_b – 3c), i.e. $[\text{CH}_3\text{COO}^-]$. Allow FT error from $[\text{OH}^-]$. Do not penalise incorrect decimal places. If 0.05 L is used for 0.065 L in 3c), then: $[\text{CH}_3\text{COO}^-] = 0.03 \text{ M}$ $[\text{OH}^-] = 4.15 \times 10^{-6}$ $\text{pOH} = 5.38 \sim 5.4$ $\text{pH} = 8.6.$

Q	Sample response	The response:	Notes
4a)	The fermentation process requires yeast as a catalyst. Yeast is temperature sensitive.	- identifies fermentation requires yeast as a catalyst [1 mark] - identifies that yeast is temperature-sensitive [1 mark]	
4b)	Cellulose is a linear polymer. The β -glucose monomers in cellulose can pack closely together. This increases hydrogen bonding between adjacent chains, which reduces interactions with water (solvents) and makes hydrolysis of cellulose more difficult than starch.	- identifies cellulose is a linear polymer [1 mark] - identifies monomers can pack closely together [1 mark] - explains increased H-bonding between adjacent chains makes hydrolysis of cellulose more difficult [1 mark]	For <i>linear polymer</i>, accept <i>not branched</i>. For <i>pack closely together</i>, accept <i>contain β-glucose monomers</i>.
4c)	$\text{Moles C}_6\text{H}_{12}\text{O}_6 = \frac{150}{180.18} = 0.833$ $\text{Moles CH}_3\text{CH}_2\text{OH} = 0.83 \times 2 = 1.67$ $\text{Mass CH}_3\text{CH}_2\text{OH} = 1.67 \times 46.08 = 76.72 \text{ g}$ $\text{Ethanol yield} = \frac{37.5}{\text{theoretical yield}} = \frac{37.5}{76.72} = 48.9\%$	- determines 150 g glucose can produce 1.67 M of ethanol [1 mark] - calculates theoretical mass of ethanol as 76.72 g [1 mark] - calculates % yield of ethanol is 48.9% [1 mark]	Do not penalise incorrect decimal places. Allow FT error from theoretical mass of ethanol.

Q	Sample response	The response:	Notes
5a)	Copper ion is reduced and Cu is plated onto the cathode: $\text{Cu}^{2+}(\text{aq}) + 2\text{e}^{-} \rightarrow \text{Cu}(\text{s})$ Copper anode is oxidised to $\text{Cu}^{2+}(\text{aq})$ and is released into solution: $\text{Cu}(\text{s}) \rightarrow \text{Cu}^{2+}(\text{aq}) + 2\text{e}^{-}$ Therefore, for every copper ion that is reduced at the cathode, in principle, another one is oxidised at the anode. Therefore, the concentration of the copper(II) sulfate solution should stay the same.	- identifies copper ions are reduced to Cu metal at the cathode and reduction half-equation is $\text{Cu}^{2+}(\text{aq}) + 2\text{e}^{-} \rightarrow \text{Cu}(\text{s})$ [1 mark] - identifies copper metal is oxidised to Cu ions at the anode and oxidation half-equation is $\text{Cu}(\text{s}) \rightarrow \text{Cu}^{2+}(\text{aq}) + 2\text{e}^{-}$ [1 mark] - predicts no change in concentration of copper(II) sulfate solution [1 mark] - identifies that copper ions are reduced to copper and copper is oxidised to copper ions at the same rate [1 mark]	States are not required in half equations. <i>For no change in concentration, accept concentration of the copper(II) sulfate solution should stay the same until anode depleted.</i>
5b)	Silver is below copper in the reactivity series and therefore doesn't go into solution, as ions are not oxidised and could be found in the sludge. Zinc impurities are above copper in the electrochemical series and will form ions at the anode and go into solution. However, they won't get discharged at the cathode, provided their concentration doesn't get too high.	- identifies Ag is less reactive than Cu and Zn is more reactive than Cu [1 mark] - deduces Ag metal is not oxidised (or reduced) and remains as metal [1 mark] - deduces Zn metal is oxidised to form ions and found in the solution [1 mark] - explains that Zn^{2+} ions remain in solution at low concentration but are reduced to Zn metal at the cathode if concentration becomes too high [1 mark]	<i>For less reactive, accept below in the electrochemical series.</i> <i>For more reactive, accept above in the electrochemical series.</i>

Q	Sample response	The response:	Notes
6a)	Peptide bond Water is formed	- identifies peptide bond [1 mark] - identifies water is formed [1 mark]	
6b)	6 tripeptides can be formed: Gly-His-Lys and Gly-Lys-His	- determines 6 tripeptides can be formed [1 mark] - describes one tripeptide [1 mark] - describes a second tripeptide [1 mark]	For <i>descriptions of a tripeptide</i>, accept: - Gly-His-Lys - Gly-Lys-His - Lys-Gly-His - His-Gly- Lys - Lys-His-Gly - His-Lys-Gly.
6c)	The isoelectric point of His is 7.6. If the pH of the buffer solution is 7.6 histidine will not migrate towards either the anode or cathode because it remains neutral. The isoelectric point Gly is 6, which is less than 7.6. Therefore, in a buffer solution of 7.6 Gly will form a negative ion and migrate towards the anode. Similarly, the isoelectric point for Lys is 9.7, therefore, Lys will form a positive ion and migrate towards the cathode.	- explains when buffer solution is pH 7.6 - histidine is neutral and will not migrate [1 mark] - Gly forms a negative ion and migrates towards anode (positive terminal) [1 mark] - Lys forms a positive ion and migrates towards the cathode (negative terminal) [1 mark]	Allow FT error for migration of ions if incorrect charge identified and they are correctly applied.



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